THE INJURED PATIENT

Give urgent attention to the injured patient:

- First assess and manage airway, breathing, circulation and level of consciousness ⮌ 14.
- Identify all injuries and look for cause: undress patient and assess front and back. If head or spine injury, use log-roll to turn. Then cover and keep warm.

Bruising and blood in urine

Give sodium chloride 0.9% 1L IV hourly for 2 hours, then 500mL hourly. Aim for urine output > 200mL/hour. Stop if breathing worsens.

Wound and any of:

- Poor perfusion (cold, pale, numb, no pulse) below injury
- Excessive or pulsatile bleeding
- Penetrating wound to head/ neck/chest/abdomen

- If BP < 90/60, give sodium chloride 0.9% 1L IV rapidly, repeat until systolic BP > 90. Continue 1L 6 hourly. Stop if breathing worsens.
- If excessive or pulsatile bleeding, apply direct pressure and elevate limb.
- If bleeding severe and persists, apply tourniquet above injury.

Fracture and any of:

- Poor perfusion (cold, pale, numb, no pulse) below fracture
- Increasing pain, muscle tightness, numbness in limb
- Suspected femur, pelvis or spine fracture
- Weak/numb below fracture
- Open fracture
- > 2 rib fractures
- Severe deformity

- If pain severe, give morphine 10mg IM or 3-10mg slow IV[1]. Avoid if severe head injury.
- If poor perfusion, weakness/numbness below fracture: gently re-align into normal position.
- If open fracture: remove foreign material, irrigate with sodium chloride 0.9% and cover with saline-soaked gauze. Give ceftriaxone 1g IV[2]/IM.
- Splint limb to immobilise joint above and below fracture.
- If pelvic fracture, tie sheet tightly around hips to immobilise.

Head injury and any of:

- Any loss of consciousness
- Seizure/fit
- Severe headache
- Amnesia
- Suspected skull fracture
- Bruising around eyes or behind ears
- Blood behind eardrum
- Blood or clear fluid leaking from nose or ear
- Pupils unequal or respond poorly to light
- Weak/numb limb/s
- Vomiting ≥ 2 times
- ≥ 1 other injury
- Drug or alcohol intoxication

- If GCS < 15, neck/spine tenderness, weak/numb limb or abnormal pupils, apply rigid neck collar and sandbags/ blocks on either side of head.
- If pupils unequal or respond poorly to light, keep body straight and tilt to raise head (avoid bending spine).
- If fits, avoid diazepam/midazolam, give phenytoin[3] 20mg/kg IV in 200mL of sodium chloride 0.9% (not dextrose) over 60 minutes.

- Refer urgently. While awaiting transport, check BP, pulse, respiratory rate, oxygen saturation and GCS every 15 minutes. If open wound, give tetanus toxoid 0.5mL IM if none in past 5 years.
- If BP < 90/60, pulse > 100 or < 50, respiratory rate > 20 or < 9, oxygen saturation < 94% or drop in GCS, reassess airway, breathing, circulation, level of consciousness ⮌ 14.

Approach to the injured patient not needing urgent attention:

- Refer same day if pregnant, known bleeding disorder, on anticoagulant, involved in high-speed collision, ejected from or hit by vehicle or fell > 3 metres. If assault or abuse ⮌ 88.
- If open wound, give tetanus toxoid 0.5mL IM if none in past 5 years.

Wound

- Apply direct pressure to stop bleeding. Remove foreign material, loose/dead skin. Wash well with chlorhexidine 0.05% aqueous solution under running water for 5 minutes. Apply povidone iodine 10% solution if dirty.
- If sutures needed: inject lidocaine 1% or 2% 3mg/kg[5] around wound to numb area. Apply non-adherent dressing for 24 hours.
- Avoid suturing if > 12 hours (body), > 24 hours (head/neck), remaining foreign material, infected, gunshot or deep puncture:
  - If not suitable for suturing: pack wound with saline-soaked gauze and give cefalexin[6] 500mg 6 hourly for 5 days.
  - Review in 2 days. Suture if needed and no infection unless gunshot/deep puncture (irrigate and dress every 2 days instead).
- Give paracetamol 1g 4-6 hourly (up to 4g in 24 hours) as needed for up to 5 days.
- Advise patient to return if signs of infection (red, warm, painful, swollen, foul-smell or pus).
- Remove sutures after 5 days (face), 4 days (neck), 10 days (leg) or 7 days (rest of body).
- Refer if unable to close wound easily, weakness/numbness below injury or cosmetic concerns.
- If not healed completely after 3 months or heals by < 50% after 6 weeks on treatment →76.

Fracture

- Splint limb to immobilise joint above and below fracture.
- Give paracetamol 1g 4-6 hourly (up to 4g in 24 hours) and add ibuprofen[7] 400mg 8 hourly with food for up to 5 days if needed.
- Do x-ray and refer to doctor same day.

Head injury

- Observe for 2 hours before discharging.
- If mild headache, dizziness or mental fogginess, **concussion** likely:
  - Advise complete rest for 2 days. If no symptoms ≥ 3 days, gradually increase exertion.
  - Advise that recovery can take > 1 month.
  - Give paracetamol 1g 4-6 hourly (up to 4g in 24 hours) as needed for up to 5 days.
- Advise to return immediately if any of above symptoms of severity develop.

[1]Dilute 10mg morphine with 9mL of sodium chloride 0.9%. Give diluted morphine 3mL IV over 3 minutes (1mL/minute). If needed, give another 1mL/min until pain improved, up to 10mL. Stop if BP drops < 90/60. [2]Do not mix Ringer's lactate and IV ceftriaxone. Flush IV line with sodium chloride 0.9% before and after IV ceftriaxone. [3]IV phenytoin can cause low blood pressure and heart dysrhythmia: maximum infusion rate is 50mg/minute; monitor ECG and BP. If IV phenytoin unavailable, give face mask oxygen and refer urgently. [4]One drink is 1 tot of spirits, or 1 small glass (125mL) of wine or 1 can/bottle (330mL) of beer. [5]To calculate volume to inject, use 0.15mL/kg of lidocaine 2% and 0.3mL/kg of lidocaine 1%. [6]If cefalexin unavailable, use instead flucloxacillin 500mg 6 hourly for 5 days. If severe penicillin allergy (history of anaphylaxis, urticaria or angioedema), give azithromycin 500mg daily for 3 days instead. [7]Avoid ibuprofen if peptic ulcer, asthma, hypertension, heart failure, kidney disease.